



**NUS**  
National University  
of Singapore

**What2Watch**  
**BT3103 Final Report**  
**Project Group: L2 16**

| <b>Name</b>          | <b>Student No.</b> |
|----------------------|--------------------|
| Li Han Lin           | A0273744X          |
| Ng Zheng Leng Bryan  | A0273387R          |
| Heng Hong Quan       | A0273589J          |
| Jeffrey Peh Jun Ting | A0277580U          |
| Tai Hong Wei         | A0277484M          |

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# 1 Introduction

## 1.1 Executive Summary

The rise of streaming platforms has transformed the way people consume entertainment, offering an enormous and wide amount of content. However, with an overwhelming number of new streaming platforms each with its own exclusive content and heavy marketing from the major streaming companies, viewers often struggle with decision paralysis or over-subscribing to multiple streaming services but not fully utilizing them, leading to unnecessary spending on these additional streaming platforms (Holland, 2021). Mainstream productions dominate the spotlight having more resources to promote their films, leaving lesser-known films and independent productions overshadowed (Hassenger et al., 2024).

Our web application aims to address these challenges by providing a centralized platform where users can efficiently explore and compare content across multiple streaming services. By providing an objective and unbiased overview of available platforms, users can make informed decisions about their subscriptions rather than feeling compelled to sign up for multiple services unnecessarily. The application would allow users to pick and choose their favourite movies, after which it generates a list of streaming platforms that offer the highest number of their chosen movies. The application would help users make more informed decisions as to which streaming platforms to subscribe to while minimizing any underutilized spending.

Beyond assisting with streaming services recommendations, our platform is designed to highlight lesser-known films that are not heavily marketed but offer unique storytelling and creative originality. If users decide to explore lesser-known films during the onboarding process, users will receive tailored recommendations based on their preferences. The application will then suggest the most suitable indie films that align with their preferences and streaming platforms they can view these films on, helping the users discover unique indie films that they might enjoy.

Our web application aims to provide a seamless and personalized experience for users struggling to find the right streaming platform or discover films that align with their interests. By simplifying content discovery and consolidating streaming options, we help users make informed decisions while minimizing the complications when deciding on a streaming platform. Whether they are searching for timeless classics or exploring hidden gems, our platform ensures a tailored and enjoyable viewing experience for individuals, families, and friends alike.

## 1.2 Problem Statement

In today's entertainment industry, the rise of streaming platforms has revolutionized the way people consume content. With an overwhelming number of services like Netflix, Disney+, and countless others, consumers are faced with an abundance of choices (Holland, 2021). While this variety offers access to movies and series from all over the world, it also creates a paradox of choice, a situation where having too many options leads to decision paralysis (Cottrell, 2024). Instead of easily picking a show to watch, viewers often spend more time scrolling through endless catalogues, struggling to make a decision. This hesitation can lead to frustration, wasted time, and even a decline in engagement, as the sheer number of possibilities becomes mentally exhausting rather than enjoyable. As the content landscape continues to expand, finding a way to simplify choices and enhance user experience has become a growing necessity.

Streaming platforms are constantly competing for audience attention, heavily advertising their latest shows and blockbuster releases across social media, billboards, and digital platforms. This aggressive marketing strategy prioritizes high-budget productions and popular franchises, making it increasingly difficult for lesser-known films and independent productions to gain visibility. As a result, many hidden gems—films with unique storytelling, diverse perspectives, and creative originality—struggle to find an audience. Viewers, bombarded with the most hyped-up content, often miss out on films that might align more closely with their personal tastes, simply because they are not given the same level of exposure (Hassenger et al., 2024). This creates a biased entertainment landscape where only the mainstream content gets any exposure, while equally valuable but lesser-known productions remain overshadowed.

Beyond the challenge of selecting what to watch, consumers also face the dilemma of choosing which streaming services to subscribe to. With each platform offering exclusive content, viewers often feel pressured to sign up for multiple services just to access a handful of shows they are interested in. However, as new platforms continue to emerge and subscription costs add up, many end up oversubscribing, paying for services they rarely use (Ford, 2024). This leads to subscription fatigue, where users are burdened by the financial and mental load of managing multiple accounts, often without fully utilizing them.

## 1.3 Purpose

The purpose of our web application is to provide a centralized platform where users can easily explore and compare the programming offered by different streaming services. By consolidating content from multiple platforms in one place, our application helps users make informed decisions about which streaming services to subscribe to, ensuring they get the most value for their entertainment budget. Additionally, our platform seeks to

highlight lesser-known films, giving them the visibility they often lack in a market dominated by heavily advertised blockbusters. Instead of relying solely on critics' opinions, viewers will have the ability to rank, review, and evaluate films, fostering a community-driven approach to movie recommendations.

Our application aims to give an objective view of all streaming content to reduce decision paralysis, helping users find content that matches their interests without feeling overwhelmed by excessive choices. This not only enhances the viewing experience but also prevents unnecessary overspending on underutilized subscriptions, allowing users to manage their entertainment expenses more effectively. By bridging the gap between content discovery and informed decision-making, our application empowers users to take control of their streaming experience, ensuring they get the most enjoyment and value from their subscriptions.

## 1.4 Scope

Our web application is designed to help users discover lesser-known films that are often overshadowed by mainstream media and heavy advertising. By providing a platform for underrepresented movies, we aim to showcase unique films that might otherwise go unnoticed. Users can search for movies, create a personalized watchlist, and receive tailored recommendations based on their preferences. Additionally, users can create a personalized watchlist of movies and shows they are excited about, and the web application will recommend the best streaming platform that offers most of their selected content. This feature helps users minimize the number of subscriptions needed, ultimately saving money while reducing decision paralysis when choosing both a streaming service and what to watch.

# 2 Objectives

## 2.1 User Benefits

### **Personalized Movie Discovery**

Users can enjoy a personalized browsing experience that recommends movies based on their interests in specific genres, origins, or other criteria. This tailored approach helps users discover movies that align with their tastes and preferences, enhancing their viewing experience.

### **Efficient and Intuitive Navigation**

The website offers an intuitive and user-friendly interface that allows users to navigate through a vast database of movies easily. Advanced search and filtering options enable

users to quickly find movies that meet specific criteria, saving time and making the movie selection process more efficient.

### **Detailed Movie Insights**

Each movie's profile page provides comprehensive details such as synopsis, cast, trailer links, streaming platforms, and ratings. This information helps users make informed decisions about what to watch next.

### **Watchlist Feature**

Allows users to save movies to a personalized watchlist, making it easy to keep track of films they wish to watch later. This feature enhances user convenience by maintaining a curated list of movies that are accessible anytime for future viewing.

### **Cost Saving**

In the Watchlist Page, a feature analyzes the movies users have added to their watchlist to display the distribution across various streaming platforms such as Netflix, Disney+, Amazon Prime, and Apple TV. This functionality helps users identify which streaming service hosts the majority of their preferred movies, allowing them to make more informed decisions about which subscriptions are most cost-effective based on their viewing preferences.

## **2.2 User Stories**

### **User Authentication**

1. As a new user, I want to create an account so I can save my preferences and receive tailored movie recommendations.
2. As a user, I want to log in securely so I can access my personalized content.
3. As a user, I want to log out of my account to protect my personal data and privacy.

### **Forgot Password**

4. As a user, I want to reset my password if I forget it, so I can regain access to my account.

### **Onboarding Questionnaire**

5. As a new user, I want to complete an onboarding questionnaire to input my genre preferences and interests, so I receive relevant movie suggestions.

## **Movie Recommendation**

6. As a user, I want to see popular and trending movie suggestions so I can discover what others are watching.

## **Recent Movies**

7. As a user, I want to view my recent activity so I can keep track of what I've searched or watched.

## **Watchlist**

8. As a user, I want to save movies to a watchlist so I can revisit them easily later.
9. As a user, I want to remove movies from my watchlist once I've watched or lost interest in them.

## **Search**

10. As a user, I want to search for movies using keywords so I can quickly find titles I'm interested in.
11. As a guest, I want to search and explore movies without signing up so I can get a feel of the app before creating an account.
12. As a user, I want to click on a movie from the search or recommendation results to view its detailed information, so I can learn more about it before watching.

## **Filtering Movies**

13. As a user, I want to limit the number of search results returned so the list is manageable and relevant.
14. As a user, I want to filter movies by genre, release year, or rating so I can find movies that match specific criteria.
15. As a user, I want the filtering and sorting of movie lists (e.g., by rating or Title) to work properly, so I can explore content in meaningful ways.

## **Movie Streaming Distribution**

16. As a user, I want to track my watchlist and see which streaming platforms host those movies.

### **About Us & Contact Us**

- 17. As a user, I want to access a "Contact Us" page to provide feedback or request support when needed.
- 18. As a user, I want to access the "About Us" page easily from the navigation menu, so I can find it when I need to know more about the app.

### **Profile Management & Settings**

- 19. As a user, I want to view and update my movie preferences in the Profile section to receive more accurate recommendations.
- 20. As a user, I want to change my password securely from the Settings tab to maintain account security.
- 21. As a user, I want my login session to persist as I navigate through the app, so I don't need to re-authenticate on every page.
- 22. As a user, I want to navigate smoothly across different views (Dashboard, Guest, Watchlist, Recents, etc.), so I can access various features of the app.

## **2.3 Database Structure**

The What2Watch app uses a streamlined Firestore NoSQL database to deliver fast, scalable, and personalized movie recommendations. Each user is represented as a document in the Users collection, storing basic info like email, preferences (genres, languages, countries), and timestamps. Two subcollections — Recents and Watchlist — store recently viewed and saved movies as documents with metadata such as title, genres, providers, and rating. This structure supports features like activity tracking and streaming platform summaries. By storing only user-specific data and fetching movie details from the TMDb API, the setup ensures efficient querying, real-time updates, and a responsive user experience.

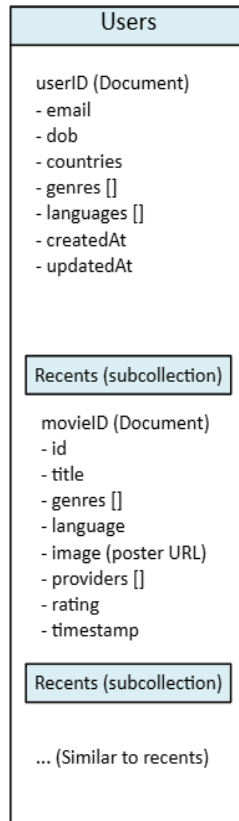


Fig 1: Document Database

## 2.4 Data Flow Diagram

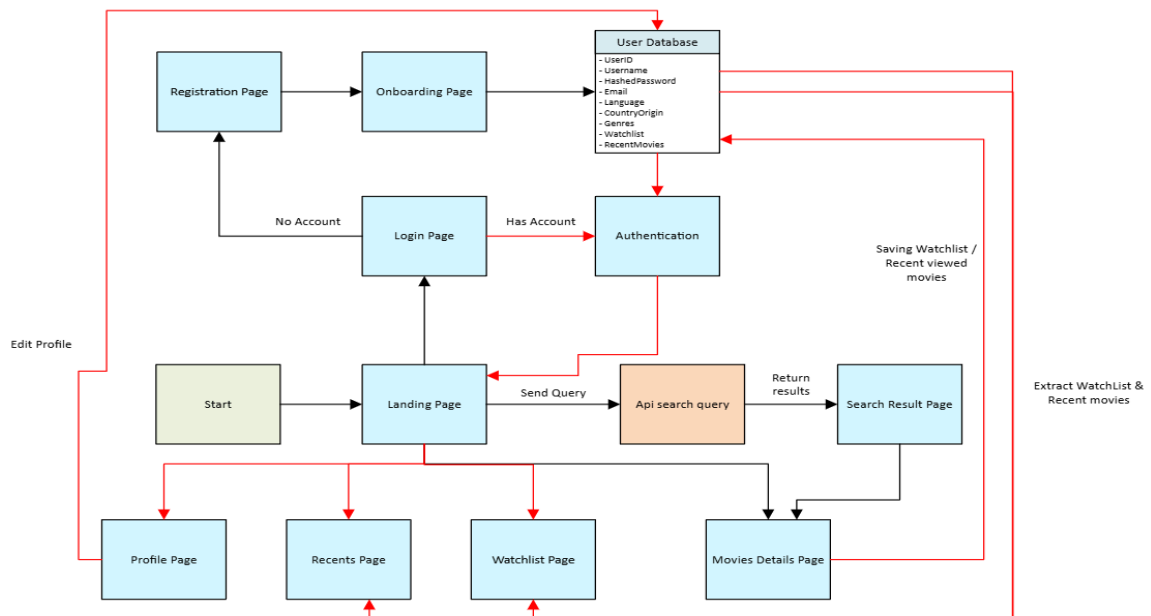


Fig 2: Data Flow Diagram

The data flow diagram in Fig. 2 outlines our movie recommendation application's structure and user interactions. Starting on the landing page, new users input preferences like favorite genres and film types during onboarding, with this data stored in the User Database (Fig. 1). Users can opt to browse without an account, accessing only specific features linked by black arrows, where the features linked by red arrows require authentications.

Post-registration, the authentication process secures user login, leading them to the Landing Page. Here, personalized movie data is dynamically fetched via API based on user preferences, optimizing the content delivery and system efficiency.

Users can edit personal details on their Profile Page or manage their saved movies on the Watchlist Page, directly enhancing engagement by facilitating easy access to favored films. The Search Results Page displays movies matching user searches, encouraging further exploration. Additionally, the Recents Page shows a list of recently viewed or searched movies, providing users with a convenient way to revisit or resume exploring films they've shown interest in.

Further interactions occur on the Movie Details Page, where detailed information like cast, crew, synopsis, and trailers are fetched when a movie is selected. This ensures current information without the need for extensive local data storage, supporting our goal of maintaining a streamlined and efficient database architecture.

## 2.5 User Flow Diagram

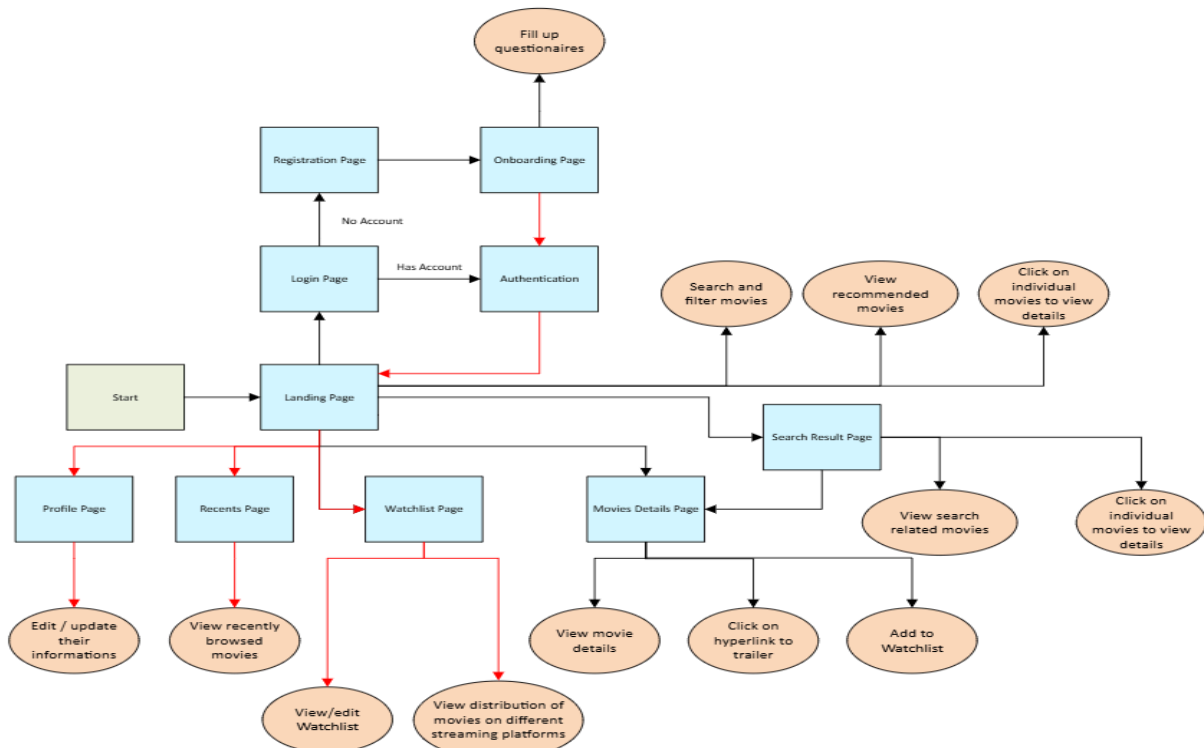


Fig 3: User Flow Diagram

The diagram in Fig. 3 outlines the user journey within our movie recommendation application, starting with the Registration Page where new users sign up and complete questionnaires on the Onboarding Page to personalize their experience based on movie preferences. Existing users access the platform through the Login Page, ensuring secure entry.

Once logged in, users arrive at the Landing Page, the application's central hub, which displays personalized movie recommendations and offers search tools to refine movie selections. Users can navigate to the Profile Page to edit their details, to the Watchlist Page to review or manage their watchlist, and to the Recents Page to view a list of recently viewed or searched movies.

Further, the Movies Details Page accessible from the Landing Page and Search Result Page provides extensive information about each movie, including synopses, cast, and hyperlinks to watch trailers directly. This streamlined flow enhances user engagement by providing a seamless navigation experience and immediate access to detailed movie content.

## 3 Requirement Analysis

### 3.1 Functional Requirements

| S/N                                       | Requirements Description                                                                                                                              | Acceptance Criteria                                                                                                                                                        | Priority |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Landing Page (Pre-Login Dashboard)</b> |                                                                                                                                                       |                                                                                                                                                                            |          |
| FR-LAN-001                                | <b>Movie Recommendations</b><br><br>Users should see a list of recommended (highly rated) movies.                                                     | The system must display at least 10 recommended movies per user session, based on movie ratings.                                                                           | High     |
| FR-LAN-002                                | <b>Authentication</b><br><br>Users should be able to perform authentication to access personalized features.                                          | The system must provide a button to route to the user authentication page.<br><br>The user must be restricted from personalized features unless the user is authenticated. | High     |
| FR-LAN-003                                | <b>Search with Filter</b><br><br>Users should be able to search for movies with string input, filtering fields of their choice.                       | The search with filter function must be accessible and return relevant results within 5 seconds based on selected filters.                                                 | High     |
| FR-LAN-004                                | <b>About</b><br><br>Users should be able to see an “About” section containing platform details.                                                       | The “About” must be accessible from the main navigation bar.<br><br>The platform details must be displayed correctly.                                                      | Low      |
| FR-LAN-005                                | <b>Contact Us</b><br><br>Users should be able to see a “Contact Us” section containing developers’ contact details.                                   | The “Contact Us” must be accessible from the main navigation bar.<br><br>The developers’ contact details must be displayed correctly.                                      | Low      |
| <b>Authentication</b>                     |                                                                                                                                                       |                                                                                                                                                                            |          |
| FR-AUT-001                                | <b>User Registration</b><br><br>Users should be able to create an account using a valid email and password. Emails already registered with an account | The system must send users a success message after successful user registration.<br><br>The system must save the user name, email, and password into                       | High     |

|                   |                                                                                                                 |                                                                                                                                                                                                      |        |
|-------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                   | should not be allowed for registration.                                                                         | the database.<br><br>Otherwise, an error message should be displayed.                                                                                                                                |        |
| FR-AUT-002        | <b>User Login</b><br><br>Users should be able to log in using their registered credentials.                     | The system must validate login credentials and allow access only if correct.<br><br>Upon successful login, direct the user to the dashboard.<br><br>Incorrect attempts should show an error message. | High   |
| FR-AUT-003        | <b>User Logout</b><br><br>Users should be able to log out of their accounts.                                    | The system must display a message upon successful logout, and direct the user to the guest landing page.                                                                                             | High   |
| FR-AUT-004        | <b>Password Reset</b><br><br>Users should be able to reset their password if they forget it.                    | The system must prompt users to input their current password again when attempting to reset their password.                                                                                          | Medium |
| <b>Onboarding</b> |                                                                                                                 |                                                                                                                                                                                                      |        |
| FR-ONB-001        | <b>Questionnaire</b><br><br>Users should be able to set up their profile with a questionnaire upon first login. | The system must display a questionnaire upon first login.<br><br>The questionnaire should display a set of questions for them to fill in and confirm or skip.                                        | High   |
| FR-ONB-002        | <b>Preferred Language</b><br><br>Users should be able to specify their preferred language(s) for content.       | The system should allow users to select multiple languages and the system must use this data to refine recommendations.<br><br>This data must be stored in the database.                             | Medium |
| FR-ONB-003        | <b>Preferred Countries of Origin</b><br><br>Users should be able to choose their preference for countries of    | The system should allow users to select countries of origin of movies from at least 10 countries and must use this                                                                                   | Medium |

|                  |                                                                                                                |                                                                                                                                                                                                 |        |
|------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                  | origin of movies                                                                                               | data to refine recommendations.<br><br>This data must be stored in the database.                                                                                                                |        |
| FR-ONB-004       | <b>Preferred Genre</b><br><br>Users should be able to choose their favorite genres for better recommendations. | Users should be able to pick multiple genres, and the system must use this data to refine recommendations.<br><br>This data must be stored in the database.                                     | Medium |
| FR-ONB-005       | <b>Birth Date</b><br><br>Users should input their birth date for content filtering purposes.                   | The system should allow users to enter their age and the system must use this data to restrict content if necessary.                                                                            | Medium |
| <b>Dashboard</b> |                                                                                                                |                                                                                                                                                                                                 |        |
| FR-DAS-001       | <b>Recents</b><br><br>Users should be able to view a list of movies they recently clicked on.                  | The dashboard must display a list of the last 10 movies the user has clicked into.<br><br>A history of the movies the user clicked on should be kept in the database.                           | High   |
| FR-DAS-002       | <b>Watchlist</b><br><br>Users should be able to save movies and shows for later viewing.                       | The dashboard must display the last 10 movies the user added to their watchlist.<br><br>Users must be able to add or remove content from their watchlist, which should persist across sessions. | High   |
| FR-DAS-003       | <b>Recommended Movies</b><br><br>Users should receive personalized recommendations on the home page.           | The system must generate and display at least 10 recommendations, based on a weighted calculation of relevance to user preferences and movie ratings.                                           | Medium |
| FR-DAS-004       | <b>Discover</b><br><br>Users should be able to explore                                                         | The system must generate and display at least 10 movies that do not align with their                                                                                                            | Medium |

|               |                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                           |      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|               | high-rated movies that do not align with their preferences.                                                                                                              | preferences.                                                                                                                                                                                                                                                                                                                              |      |
| FR-DAS-005    | <b>Profile</b><br><br>Users should be able to access their profile page.                                                                                                 | The system must provide a button that links to their profile page.                                                                                                                                                                                                                                                                        | High |
| FR-DAS-006    | <b>Search with Filter</b><br><br>Users should be able to search for movies with string input, filtering fields of their choice.                                          | The search with filter function must be accessible and return relevant results within 5 seconds based on selected filters.                                                                                                                                                                                                                | High |
| FR-DAS-007    | <b>About</b><br><br>Users should be able to access a clearly defined About page that communicates the platform's mission, core features, and data sources.               | The "About" must be accessible from the main navigation bar.<br><br>The About page should display the platform's key offerings such as recommendation logic, streaming platform integration and data aggregation, a short summary of how user preferences are processed and matched, and lastly attribution to TMDb as the data provider. | Low  |
| FR-DAS-008    | <b>Contact Us</b><br><br>Users should be able to reach out to the developers through a Contact Us section that provides communication channels and availability details. | The "Contact Us" must be accessible from the main navigation bar.<br><br>The developers' contact details such as a support email address, a contact phone number, a physical address of the team and operating hours, must be displayed correctly.                                                                                        | Low  |
| <b>Search</b> |                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                           |      |
| FR-SER-001    | <b>Search Function</b><br><br>Users should be able to search for the movies that they want, based on the title                                                           | The system must take in a string input and display the corresponding results in 5 seconds.<br><br>If logged in, the returned results should be ranked by a weighted score that is calculated by relevance to user profile and                                                                                                             | High |

|                      |                                                                                                                                                             |                                                                                                                                                                                                                  |        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                      |                                                                                                                                                             | <p>movie rating.</p> <p>If no movies are found based on the filter, display “No movies are found”.</p>                                                                                                           |        |
| FR-SER-002           | <p><b>Filter</b></p> <p>Users should be able to filter movies by genre, release year, and other criteria for easy discovery.</p>                            | <p>The system should display movies according to the filters selected.</p> <p>Filters should be applicable in combination.</p> <p>If no movies are found based on the filter, display “No movies are found”.</p> | High   |
| <b>Movie Details</b> |                                                                                                                                                             |                                                                                                                                                                                                                  |        |
| FR-MOD-001           | <p><b>Movie Details</b></p> <p>Users should be able to see the details of the movies, including title, description, genre, duration, ratings, platforms</p> | The system must display all the details of the movies, including title, description, genre, duration, ratings, and platforms.                                                                                    | High   |
| FR-MOD-002           | <p><b>Watch Trailer</b></p> <p>Users should be able to watch the trailer of the movie</p>                                                                   | The system must be able to redirect the users using the correct link for the trailer.                                                                                                                            | Low    |
| FR-MOD-003           | <p><b>Add to Watchlist</b></p> <p>Users should be able to add the movie to the watchlist</p>                                                                | The system must add the corresponding movie to the watchlist and update the database.                                                                                                                            | Low    |
| <b>Watchlist</b>     |                                                                                                                                                             |                                                                                                                                                                                                                  |        |
| FR-WAT-001           | <p><b>View Watchlist</b></p> <p>Users should be able to see a list of movies under the watchlist</p>                                                        | The system must be able to display a list of movies that are placed under the watchlist by the user.                                                                                                             | High   |
| FR-WAT-002           | <p><b>Add movies to the Watchlist</b></p> <p>Users should be able to add movies to the watchlist</p>                                                        | The system must be able to add a movie to the watchlist, and the movie should appear inside the watchlist.                                                                                                       | Medium |
| FR-WAT-003           | <p><b>Remove movie from Watchlist</b></p> <p>Users should be able to remove</p>                                                                             | The system must be able to remove a movie from the watchlist, and the movie should                                                                                                                               | Medium |

|                |                                                                                                                                                                                |                                                                                                                                                                                                                            |        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                | movies from the watchlist                                                                                                                                                      | disappear from the watchlist.                                                                                                                                                                                              |        |
| FR-WAT-04      | <b>Sort movies in Watchlist</b><br><br>Users should be able to sort the movies in the watchlist according to the various filters                                               | The system must display the list of movies according to the filter selected, eg. date added (newest), date added (oldest), and most popular.                                                                               | Medium |
| FR-WAT-04      | <b>Streaming Distribution in Watchlist</b><br><br>Users should be able to see the streaming distribution of their watchlist on different streaming platform                    | The system must display the summary table indicating the counts of movies saved in watchlist in each of the streaming platform.                                                                                            | Medium |
| <b>Profile</b> |                                                                                                                                                                                |                                                                                                                                                                                                                            |        |
| FR-PRO-001     | <b>View Profile</b><br><br>Users should be able to view their personal information                                                                                             | The system must be able to display users the correct personal information clearly.                                                                                                                                         | Medium |
| FR-PRO-002     | <b>Edit Preferences</b><br><br>Users should be able to give information about their favourite genre and artists for movies through a questionnaire, and their personal details | The system must be able to prompt questions for the users regarding their favourite genre and artists for movies.<br><br>Users should be able to enter their personal information and the system must update the database. | Medium |
| FR-PRO-003     | <b>Edit Settings</b><br><br>Users should be able to amend their personal details.                                                                                              | Users can change their name, email, password, etc and the system must update the database.                                                                                                                                 | Medium |

## 3.2 Non-Functional Requirements

| S/N                | Requirements Description                                                                                                           | Acceptance Criteria                                                                                                                            | Priority |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Performance</b> |                                                                                                                                    |                                                                                                                                                |          |
| NF-REQ-001         | <b>Response Time</b><br>Application should have a quick response time for all user interactions.                                   | Web pages should load within 2.5 seconds, especially for key features.                                                                         | High     |
| NF-REQ-002         | <b>Scalability</b><br>Application should accommodate increasing user traffic without causing it to crash.                          | The system should support at least 30 concurrent users without performance degradation.                                                        | High     |
| NF-REQ-003         | <b>Throughput</b><br>Application should handle high volumes of API requests                                                        | Application should handle at least 30 requests per second.                                                                                     | High     |
| <b>Security</b>    |                                                                                                                                    |                                                                                                                                                |          |
| NF-REQ-004         | <b>Data Encryption</b><br>Confidential information should be encrypted and kept secured while being transmitted and stored.        | Users' personal information and sensitive data must be encrypted.<br><br>All data must be transmitted over HTTPS to ensure secure transmission | High     |
| NF-REQ-005         | <b>Authentication &amp; Authorisation</b><br>Secure access control for users and administrators                                    | Implement JWT-based or session-based authentication for user login                                                                             | High     |
| NF-REQ-006         | <b>Audit Trail</b><br>Application should keep a log of user actions to monitor user activities and identify any suspicious actions | Maintain logs for critical actions with timestamps.<br><br>Logs should be immutable and reviewed periodically                                  | Low      |
| <b>Reliability</b> |                                                                                                                                    |                                                                                                                                                |          |
| NF-REQ-007         | <b>System Uptime</b><br>Application should ensure high availability and minimise system downtimes                                  | Application should have at least 99% uptime, monitored via health checks.<br><br>Maintenance scheduled during                                  | High     |

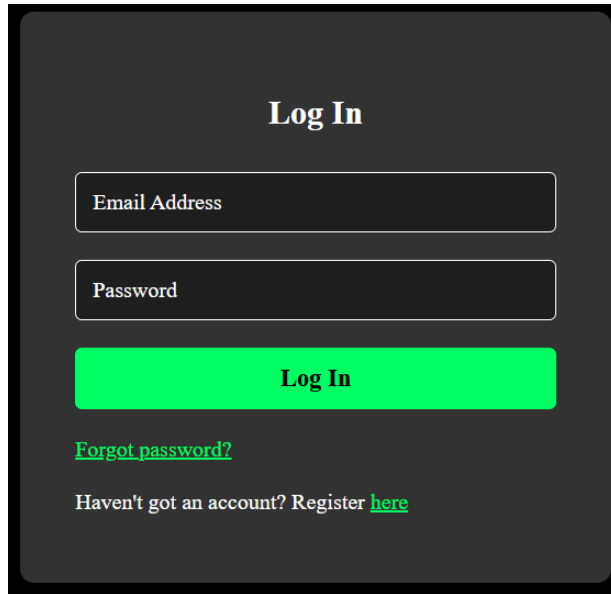
|                        |                                                                                                                    |                                                                                                                            |        |
|------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|
|                        |                                                                                                                    | off-peak hours.                                                                                                            |        |
| NF-REQ-008             | <b>Error Handling</b><br>Application should handle errors gracefully without exposing confidential information.    | Implement user-friendly error messages and HTTP status codes and log all system errors for diagnostics.                    | Medium |
| NF-REQ-009             | <b>Recovery</b><br>Recovery plans must be in place to ensure proper backup and data integrity in case of crashes.  | Daily backups of critical data and a disaster recovery plan ensuring system restoration within 3 hours of failure.         | High   |
| <b>Usability</b>       |                                                                                                                    |                                                                                                                            |        |
| NF-REQ-010             | <b>Accessibility</b><br>Application should be user-friendly to people with disabilities.                           | Comply with Web Content Accessibility Guidelines (WCAG) 2.1 standards, including color contrast and screen reader support. | Medium |
| NF-REQ-011             | <b>Device Responsiveness</b><br>Applications should adapt to different screen sizes and resolutions.               | The interface should maintain functionality on mobile, tablet and desktop devices using responsive design.                 | Medium |
| NF-REQ-012             | <b>Cross-Browser Compatibility</b><br>Application should ensure consistent experience across all major browsers.   | Application should render correctly in Chrome, Safari, Firefox, and Edge without major UI/UX discrepancies.                | Medium |
| <b>Maintainability</b> |                                                                                                                    |                                                                                                                            |        |
| NF-REQ-013             | <b>Documentation</b><br>Application should maintain comprehensive and up-to-date technical and user documentation. | Provide system architecture, API docs, deployment manuals and user guides.<br><br>Update docs with every new release.      | Medium |
| NF-REQ-014             | <b>Code Quality</b><br>Application's codebase should be clean, modular and easy to maintain.                       | Adhere to coding best practices (SOLID, DRY etc) and conduct regular code reviews                                          | High   |

## 4 Design Features

### 4.1 User Authentication

The User Authentication system has been comprehensively designed and implemented to ensure both security and usability.

#### 4.1.1 Login Functionality



**Log In**

Email Address

Password

**Log In**

[Forgot password?](#)

Haven't got an account? Register [here](#)

*Fig 4: Login Page*

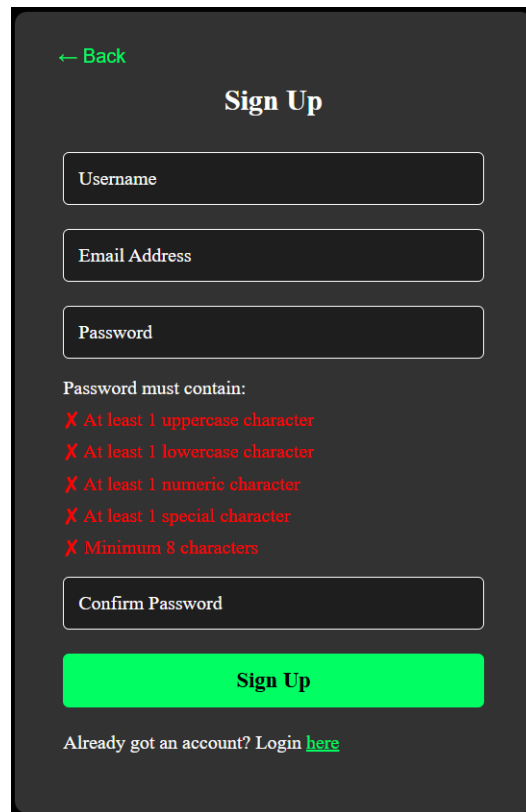
The login page is fully functional and integrates both the database and frontend components. Users can securely input their email and password to gain access to their account. The form is designed to prevent accidental submissions and includes real-time validation for empty fields and incorrect email formats. Upon submission, the credentials are securely transmitted and verified, ensuring protection against data breaches or man-in-the-middle attacks.

If the login is successful, users are redirected to their personalised dashboard or the page they previously attempted to access.

In case of incorrect credentials, a user-friendly error message is displayed to guide users without revealing specific security details (e.g., "Incorrect email or password" rather than specifying which one is wrong).

To support accessibility and navigation, the page also includes a clearly labeled forgot password link, which guides users through account recovery and a sign up link, which redirects users to the registration page.

#### 4.1.2 Registration Functionality



← Back

### Sign Up

Username

Email Address

Password

Password must contain:

- ✗ At least 1 uppercase character
- ✗ At least 1 lowercase character
- ✗ At least 1 numeric character
- ✗ At least 1 special character
- ✗ Minimum 8 characters

Confirm Password

**Sign Up**

Already got an account? Login [here](#)

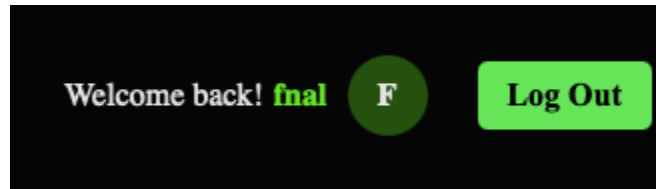
*Fig 5: Registration Page*

The registration page supports a seamless and secure onboarding process. The form collects essential information and includes client-side and server-side validation to ensure data integrity. One of the key enhancements is the real-time password validation feature, which actively assists users in creating strong passwords. As users type their password, a set of validation criteria is displayed, including: minimum of 8 characters, at least one uppercase letter, at least one lowercase letter, at least one number, and at least one special character.

Each criterion starts in red and dynamically changes to green as the condition is met. This immediate feedback not only improves user experience but also educates users on secure password practices.

These thoughtful design decisions aim to reduce friction, minimize errors during registration, and increase overall user satisfaction.

### 4.1.3 Logout Functionality

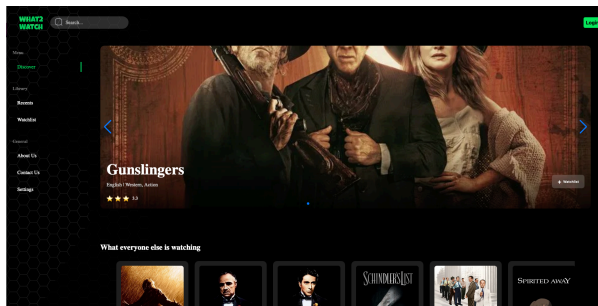


*Fig 6: Logout Functionality*

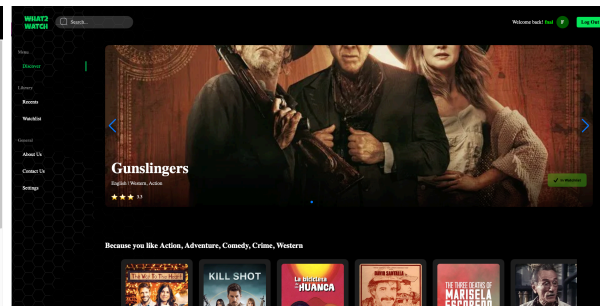
The logout functionality has been implemented to allow users to securely end their sessions. When a logged-in user clicks the “Logout” button, the application clears their authentication token from local storage and redirects them to the homepage or login page. This ensures that their session is terminated, and access to protected features like the dashboard, watchlist, and recently viewed movies is immediately revoked.

After logging out, the interface updates accordingly—hiding user-specific navigation elements and displaying only guest-accessible content. This provides a smooth and secure transition, giving users clear feedback that they are no longer signed in.

### 4.1.4 Guest Access (vs Authenticated Access)



*Fig 7: Guest Page*



*Fig 8: Dashboard Page*

The application enforces clear boundaries between guest (unauthenticated) users and logged-in users to protect user-specific features and encourage registration.

Accessible to Guests:

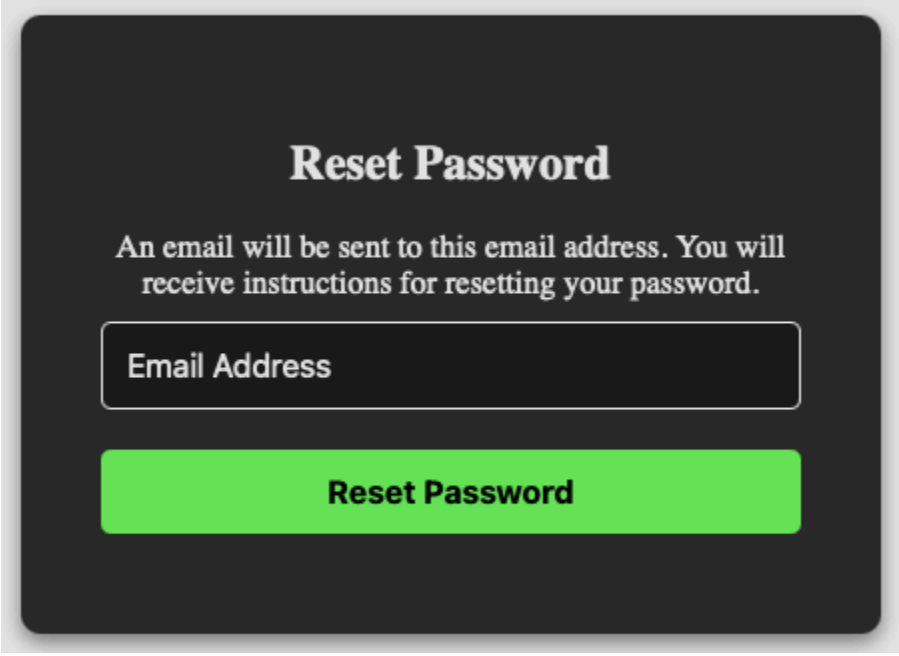
- Homepage: Guests can browse the homepage and see general information or featured content.
- Movie Search: Guests can use the search bar to look up movies by title, genre, or keyword.
- Movie Details: Clicking on a movie shows its basic information, such as title, poster, synopsis, rating, and genres.

Restricted from Guests:

- Watchlist: Guests cannot view, add to, or manage a personal watchlist.
- Recents: Only logged-in users can track and revisit their recently viewed movies.
- Personalised Recommendations Dashboard components: Guests do not have access to personalised recommendations in the dashboard

This access control is enforced both on the frontend (e.g., through route guards and conditional rendering) and backend (e.g., verifying tokens before processing requests).

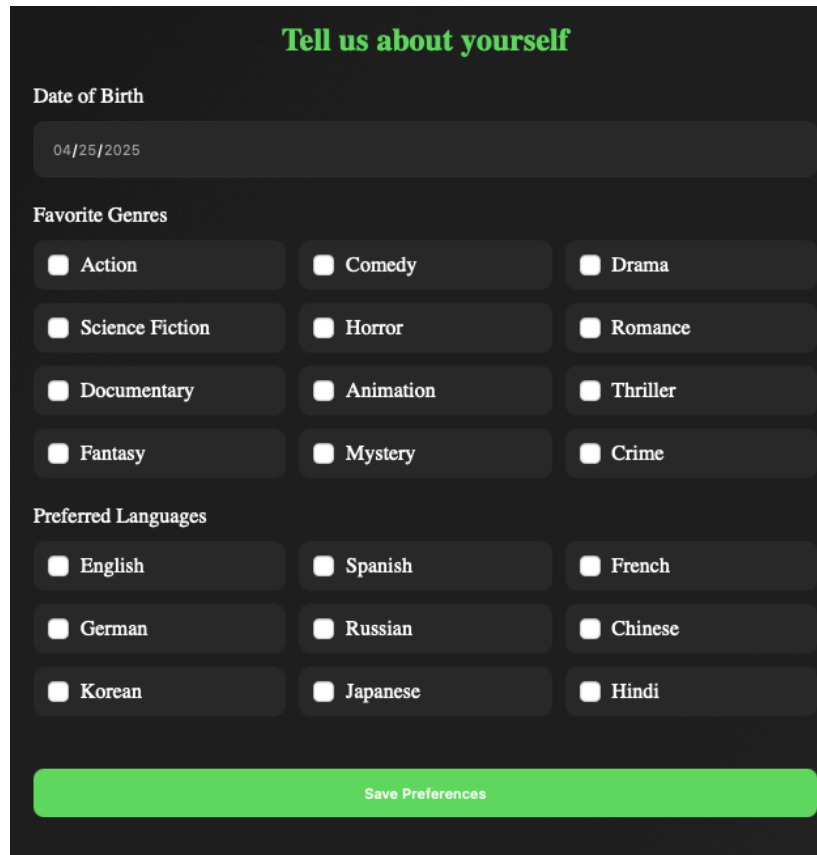
## 4.2 Forgot Password

A dark-themed modal form for resetting a password. At the top, the title "Reset Password" is centered in a bold, white font. Below the title, a message in white text states: "An email will be sent to this email address. You will receive instructions for resetting your password." Underneath this message is a white-outlined rectangular input field containing the placeholder text "Email Address". At the bottom of the form is a prominent, solid green rectangular button with the text "Reset Password" in white, bold font.

*Fig 9: Reset Password*

The “Forgot Password” feature has been fully implemented to enhance account recovery and improve user experience. Users who are unable to recall their login credentials can click on the “Forgot Password?” link on the login page, which redirects them to a password reset interface. Users will enter their registered email and the system will send a secure, one-time-use reset link for the users to reset their password

### 4.3 Onboarding Questionnaire



The image shows a dark-themed onboarding questionnaire form titled "Tell us about yourself" in green text. The form includes a "Date of Birth" section with a text input field containing "04/25/2025". Below this is a "Favorite Genres" section with a 3x4 grid of buttons, each with a white square icon and a genre name: Action, Comedy, Drama, Science Fiction, Horror, Romance, Documentary, Animation, Thriller, Fantasy, Mystery, and Crime. The last button in the grid, "Crime", is highlighted with a green background. Below the genres is a "Preferred Languages" section with a 3x3 grid of buttons, each with a white square icon and a language name: English, Spanish, French, German, Russian, Chinese, Korean, Japanese, and Hindi. The last button in the grid, "Hindi", is highlighted with a green background. At the bottom of the form is a large green button labeled "Save Preferences".

*Fig 10: Onboarding Questionnaire*

The questionnaire feature has been successfully implemented with an intuitive layout and responsive design, aimed at helping the system gather initial user preferences to improve movie recommendations. Upon first-time login or registration, users are guided to complete this onboarding form before accessing their personalized dashboard.

The form prompts users to input their Date of Birth, which may later be used to tailor age-appropriate content or improve recommendation accuracy. Following this, users are asked to select their favorite movie genres—such as Action, Comedy, Horror—and preferred languages, including options like English, German, and French. Each option is presented as a clearly labeled, toggleable button that responds with visual feedback—a green highlight—when selected. This ensures the selection process feels interactive and user-friendly.

While the current version of the questionnaire focuses solely on genres and languages, this minimal approach is intentional. It avoids overwhelming users with too many questions upfront and strikes a balance between personalization and simplicity. The

design aligns with the rest of the application by using a dark theme, paired with vibrant, consistent buttons that visually guide users through each step of the form.

Once submitted, the selected preferences are stored and used to inform the user's recommendations, improving the quality of content shown in the dashboard from the start. This onboarding step not only enhances the app's relevance but also reinforces a personalized experience from the moment the user joins.

#### 4.4 Movie Recommendation

One of the core experiences our web application offers is movie recommendation, seamlessly integrated across three distinct carousels — Popular Carousel, TasteReel, and TopRated Reel — all embedded within the main Dashboard view. These carousels collectively power a discovery experience that adapts to different levels of user engagement, ranging from anonymous exploration to fully personalized curation.



*Fig 11: Popular Carousel Component*

The Popular Carousel component acts as the most accessible form of discovery. It highlights the latest trending and widely watched movies by leveraging popularity metrics from the TMDB API. This makes it especially suitable for guest users or newcomers who haven't provided any preferences yet, ensuring they're still welcomed with appealing and up-to-date titles. The component is built using Swiper.js, a library that not only enables smooth horizontal scrolling but also introduces built-in pagination and swipe gestures — making the experience highly intuitive and mobile-friendly. Each movie card includes basic metadata, an image preview, and action buttons, offering instant access to watchlist or detail views.

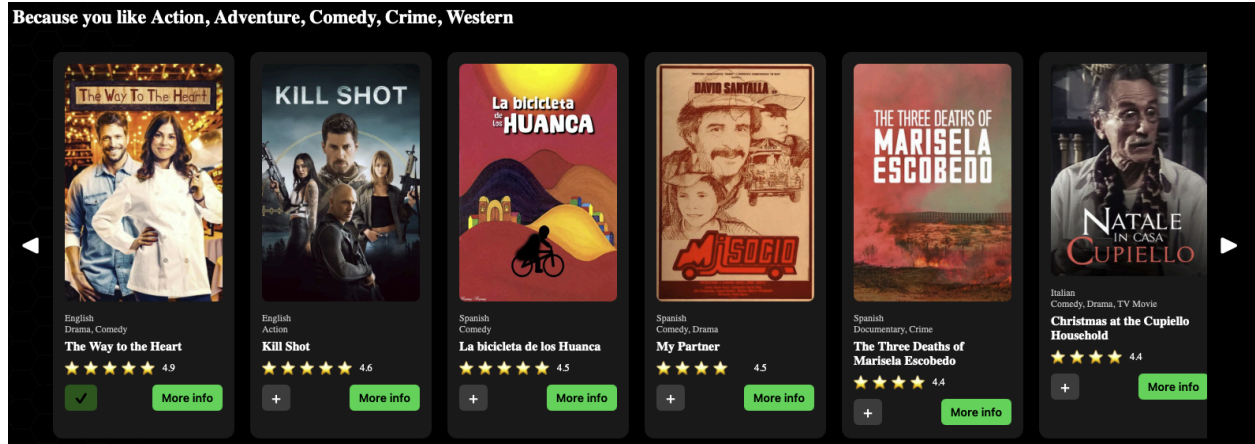


Fig 12: Taste Reel Component

The Taste Reel component is where personalization shines. This carousel recommends movies that align with a user's individual preferences — including genres, languages, countries of origin, studios, and directors — which are initially collected during onboarding and can be updated at any time from the user profile. Once preferences are set, the system maps genre names to TMDB genre IDs and retrieves a curated list of movies that match the user's taste in both genre and language. These results are then sorted by rating, ensuring only top-rated films appear in the reel. The layout mimics a film strip, allowing users to scroll through personalized selections at a glance. A distinctive feature here is the star rating system: numerical ratings out of 10 are halved and converted into a 5-star format, using full and half-star icons to provide a quick visual cue of movie quality. Additional interactions like adding to watchlist, viewing detailed info, and auto-tracking in recents are all embedded within each card, reinforcing the seamlessness of the experience.

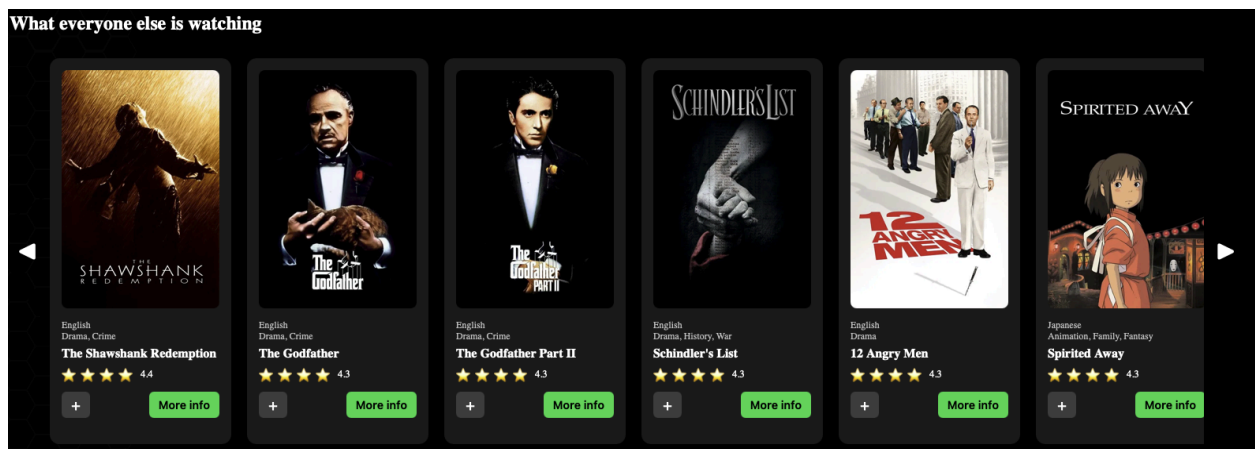


Fig 13: Top Rated Reel Component

The Top Rated Reel component completes the trio by offering a curated selection of highly acclaimed films — independent of user preferences or popularity trends. It's designed for users who seek cinematic quality and critical recognition above all else. Like the other carousels, it supports hover-based previews, clickable posters, and instant watchlist integration, but its focus remains firmly on surface-level quality: consistently high ratings, strong reviews, and timeless appeal.

Together, these three recommendation components form a robust and responsive ecosystem — one that balances mass appeal, personal relevance, and critical excellence. Whether a user is casually browsing, just getting started, or returning to explore deeper, the app offers a tailored and rewarding journey through film.

## 4.5 Recent Movies

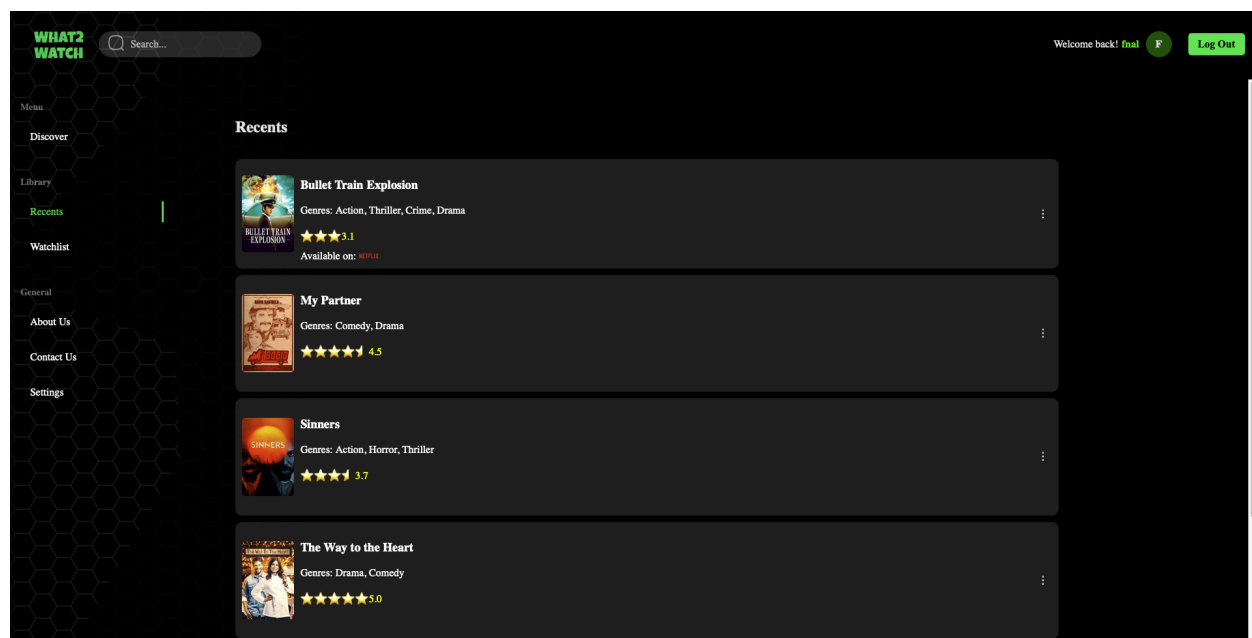


Fig 14: Recent Movies Page

To help users seamlessly continue where they left off or revisit films they've browsed, our application includes a Recents carousel displayed prominently on the homepage. This horizontally scrollable section dynamically surfaces the user's most recently clicked or viewed movies. Each time a user clicks into a movie's detail page, the title is automatically added to their personal Recents list in Firestore under the "Recents" subcollection of their user document. The carousel is personalized — meaning its contents vary by user — and is capped to display only the 50 most recent titles, ensuring relevance without overwhelming the interface. This feature enhances session continuity and encourages re-engagement with previously explored content.

## 4.6 Watchlist

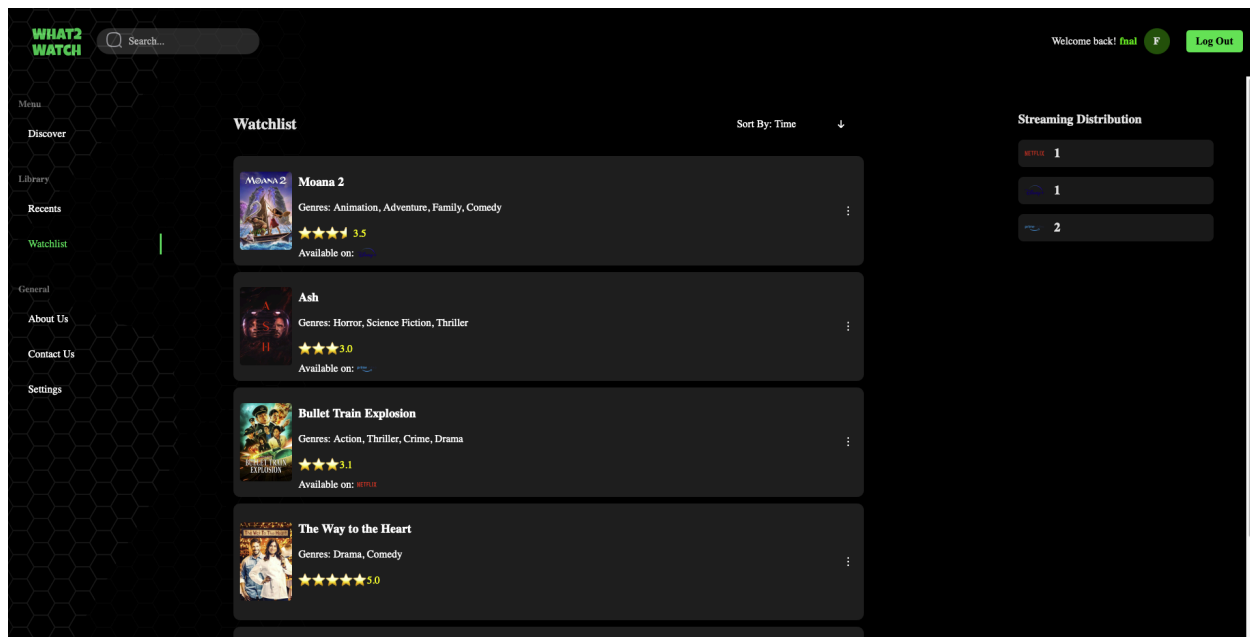


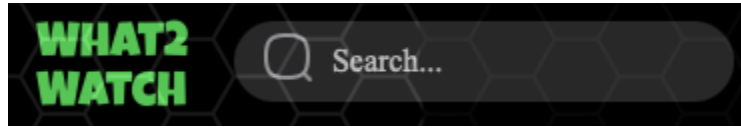
Fig 15: Watchlist Page

The Watchlist feature allows users to bookmark movies they are interested in watching, offering a personalized space for managing viewing intentions. While browsing, users can toggle movies in and out of their Watchlist using a “+” or “✓” button — these changes are reflected instantly in the UI and are stored persistently in Firebase. Specifically, each saved movie is stored in a user-specific “Watchlist” subcollection, with movie metadata saved as documents keyed by their TMDB ID. On the Watchlist page, all saved movies are displayed in a clean, scrollable layout with key details such as poster, title, genres, star rating, and streaming availability.

A sorting dropdown is available to let users sort the list by time added, offering views from newest to oldest or vice versa. Alongside the movie grid, a Streaming Distribution panel summarizes where each saved movie is currently available to stream. This data is fetched from TMDB and then counted by platform (e.g., Netflix, Prime Video), providing users with a snapshot of how many of their saved titles are accessible on services they might already subscribe to.

Together, the Watchlist and Recents features make up the core of our interaction tracking — enabling smart re-discovery, organized intent tracking, and platform-aware decision-making — all backed by real-time updates via Firebase.

## 4.7 Search



*Fig 16: Search Functionality*

The search function acts as the core discovery feature within the application. Positioned prominently on the top bar, the search function supports both keyword-based and partial match queries, allowing users to enter a search term (e.g., “Iron Man”) in the search bar. Upon submission, the system will dynamically fetch and display the relevant selection of movies in a visually engaging grid layout, with each movie represented by its poster and title. Upon clicking on a particular movie that the user is interested in, it will redirect the user to the Movie Details page, where the details of the movie will be shown

This feature significantly enhances usability by supporting targeted exploration while still encouraging users to discover different movies. It serves as a bridge between what the user is searching for and recommending other films that the user might be interested in. By adopting an intuitive input method, which is keyword search, it aligns with users expectations and habits, allowing them to quickly access the relevant movie contents without having to browse manually. This streamlined experience can improve user satisfaction and accessibility for both new and existing users.

## 4.8 Filtering Movies

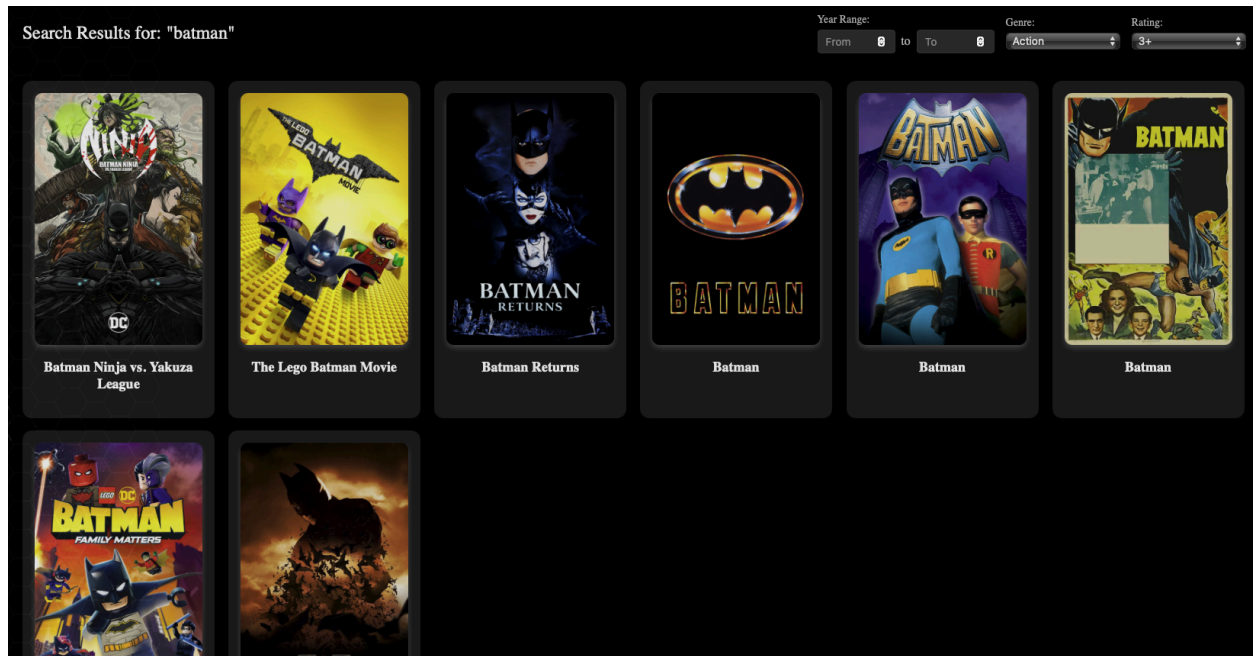
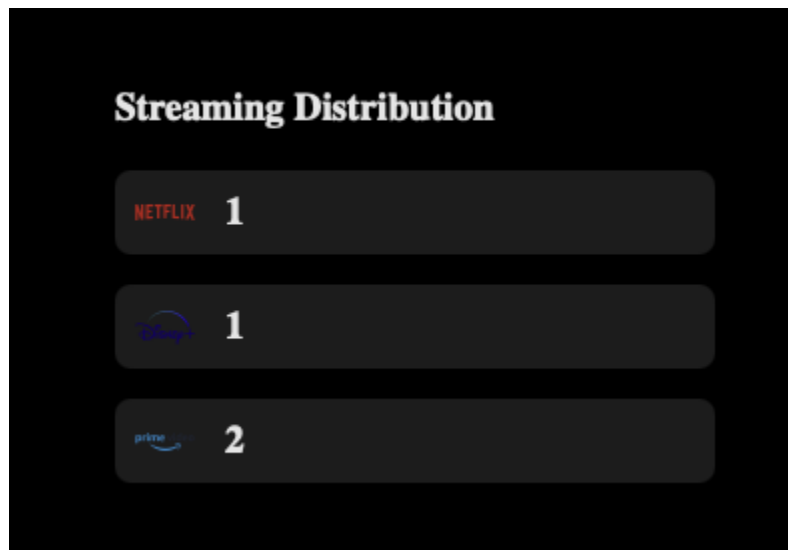


Fig 17: Filter Functionality

The filtering logic is implemented on both the watchlist and search results, ensuring real-time responsiveness and accuracy. Users can filter the watchlist based on duration, title and ratings, while users can filter search results based on released year range, genre and ratings. These filters enable users to quickly narrow down their options based on their preferences or specific criteria. The selected filters immediately update the displayed list of movies without the need for manual page refreshes, ensuring smooth and uninterrupted user experience.

This interactive and user friendly design improves the browsing and movie discovery experience by allowing for faster, more targeted navigation through large volumes of movies. It avoids information overload for the users and helps them to efficiently locate their desired content that aligns with their preferences. By reducing time and effort required to find relevant movies, the filtering function increases user satisfaction and usability of the platform, encouraging more engagement with the platform.

## 4.9 Movie Streaming Distribution



*Fig 18: Streaming Distribution Functionality*

The Movie Streaming Distribution feature enhances the user experience by providing direct access to information about where each movie can be streamed. On every movie detail page, users can view a list of available streaming platforms—such as Netflix, Disney+, Amazon Prime Video, and others—where the selected film can be watched. This integration helps users make informed decisions without needing to leave the app to search for availability elsewhere. Platform icons and labels are clearly displayed, and links redirect users to the corresponding service’s movie page in a new tab. This feature streamlines the viewing process and adds real utility by bridging the gap between discovery and consumption.

## 4.10 About Us & Contact Us

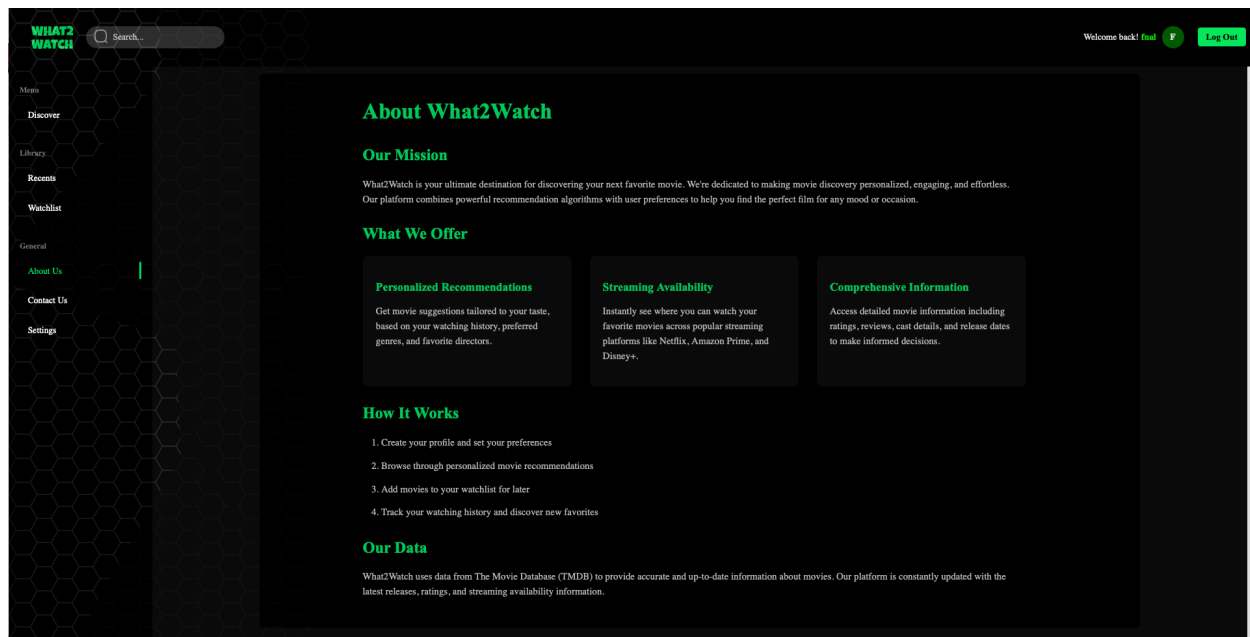


Fig 19: About Us Page

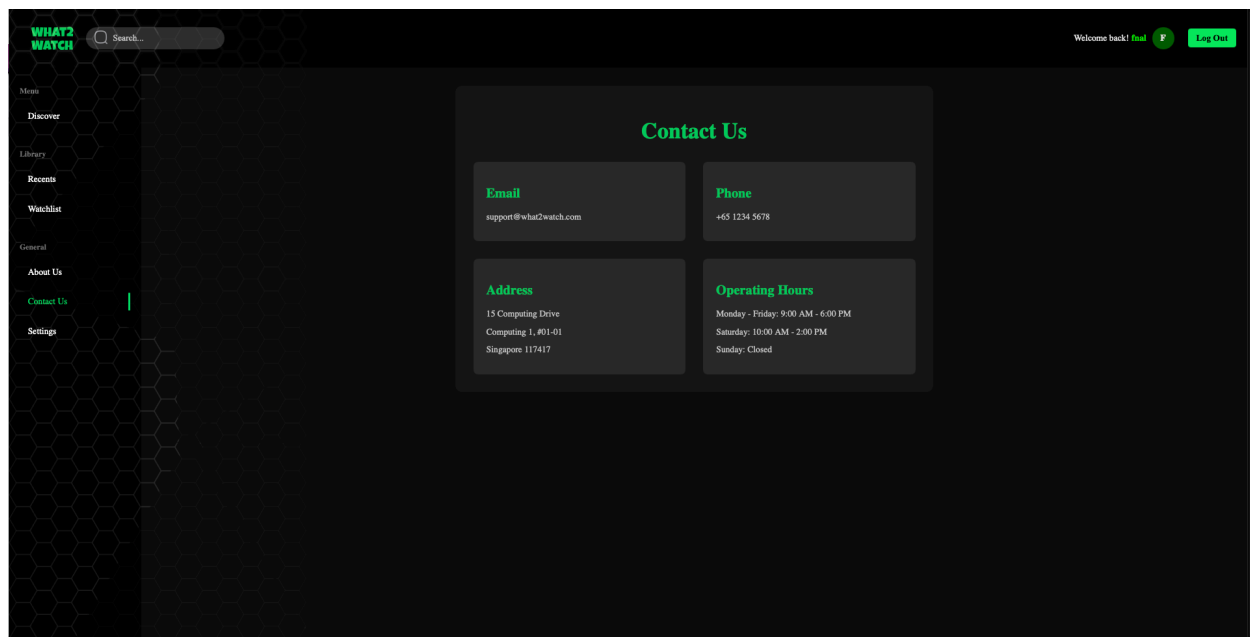


Fig 20: Contact Us Page

The About Us & Contact Us section provides users with insight into the purpose and creators behind the platform. The About Us page outlines the mission of the application—helping users navigate the overwhelming world of movie choices by offering a clean, personalised, and intelligent recommendation experience. Meanwhile,

the Contact Us page offers users a simple and responsive form to reach out with feedback, support requests, or partnership inquiries. Contact information such as email and social media links, is also included to foster transparency and encourage communication.

## 4.11 Profile Management & Settings

The screenshot shows the 'Preferences' section of the WHAT2 WATCH app. The left sidebar contains a menu with options: Menu, Discover, Library, Recents, Watchlist, General, About Us, Contact Us, and Settings (highlighted). The main content area is titled 'Preferences' and includes tabs for 'Preferences' and 'Settings'. The 'Preferences' section contains several categories of preferences, each with a set of selectable buttons:

- Preferred Languages:** EN (selected), ES, FR, DE, RU, ZH, KR, JP, HI.
- Preferred Countries of Origin:** UK, US, Spain, France, Germany, Russia, China, Korea, Japan, India.
- Preferred Movie Studios:** Marvel St..., Warner Bros., Disney, Universal, A24, Pixar, Sony Pictu..., Paramount, Netflix St..., Studio Chi...
- Favorite Directors:** Christoph..., Steven Spi..., Martin Sc..., Quentin T..., Greta Ger..., Denis Vill..., Hayao Mi..., Bong Joon...
- Preferred Genres:** Action (selected), Adventure (selected), Animation, Comedy (selected), Crime (selected), Documentary, Drama, Family, Fantasy, History, Horror, Music, Mystery, Romance, Science Fiction, TV Movie, Thriller, War, Western (selected).

Fig 21: Profile Preferences Management

The screenshot shows the 'Settings' section of the WHAT2 WATCH app. The left sidebar is the same as in Fig 21, with 'Settings' highlighted. The main content area is titled 'Settings' and includes tabs for 'Preferences' and 'Settings'. The 'Settings' section contains two main sections:

- Personal Information:** Includes fields for Username (me), Date of Birth (08/20/2018), and Email (final@gmail.com). A 'Save Info' button is located below these fields.
- Security:** Includes a 'Change Password' button.

Fig 22: Profile Settings Management

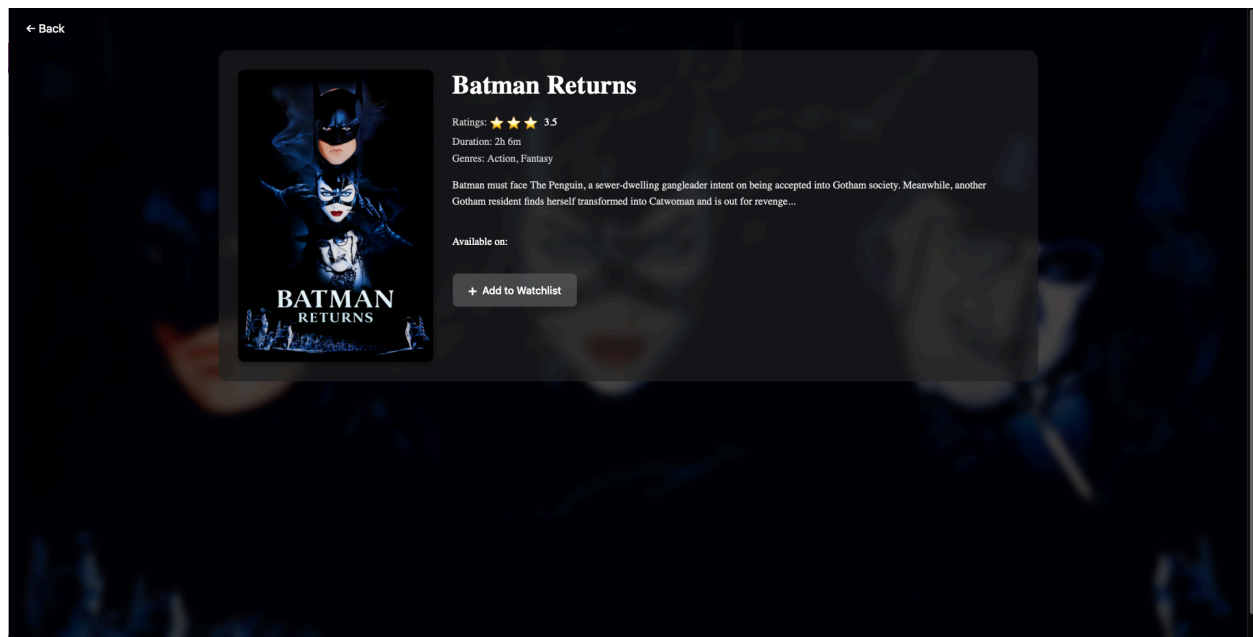
The Profile Management & Settings section empowers users to personalize their experience, maintain account information, and manage key aspects of their interaction with the platform — all within an intuitive, tabbed interface under the /profile route. This area is split into two primary tabs: Preferences and Settings, each serving distinct yet complementary functions in shaping the user experience.

The Preferences tab allows users to update their taste profile at any time after onboarding. Here, users can select and modify their preferred languages, countries of origin, movie studios, directors, and genres, alongside their typical watching frequency and preferred content length. These selections play a central role in powering the recommendation engine — especially the Taste Reel — ensuring the system evolves as user preferences change. Each category is presented in a visually appealing grid or radio-button layout, using dynamic buttons that reflect current selections with distinct color cues. The selections are synced in real time with Firestore, making updates persistent and instantly impactful across the app.

The Settings tab handles more static aspects of the user's profile, such as their username, date of birth, and email. While email remains read-only (as it is linked to Firebase Authentication), users can edit their displayed name and date of birth directly from this view. Changing the username not only updates the Firestore document but also synchronizes with Firebase Auth's displayName, ensuring consistency across components like the Topbar. Additionally, this section provides access to password management, where users can trigger a modal to change their password via Firebase's secure re-authentication flow.

All updates in this section are met with confirmation prompts or alerts, providing clear feedback on successful changes or errors. Together, the Preferences and Settings tabs offer users a high degree of control over both their content curation and account security, forming the backbone of a personalized and secure user experience.

## 4.12 Display Movie Details



*Fig 23: Movie Details Display Page*

The display movie details feature presents essential information about each movie, including the title, poster, rating, duration, and genre(s). It also includes a brief synopsis that provides an overview of the plot, giving users a quick understanding of the movie, helping them to assess whether the film aligns with their interests. If the movie appeals to the users, they can easily add it into their watchlist, saving the movie to be watched later at their convenient time. Furthermore, the movie details feature will also highlight the streaming platforms (e.g., Disney+) where the movie is available on, making it easier for users to locate the film.

This feature supports informed decision-making for the users by presenting detailed and visually appealing information about each movie. By reviewing key information such as duration, genres and ratings, users can quickly gauge whether the movie suits their mood, time and availability, reducing the need for users to conduct external searches outside the platform. The integrated watchlist feature allows users to conveniently save movies that catch their attention immediately, saving time and effort in recording their desired watchlist. Furthermore, by displaying the available streaming platforms, users can avoid the trouble of searching for the movies across multiple streaming platforms. These functionalities enhance user satisfaction and usability of the platform, making movie discovery and selection process an enjoyable and efficient experience.

## 5 SCRUM

SCRUM provided a lightweight but highly effective structure for managing our development process. By breaking our work into sprints and focusing on delivering incrementally, we were able to maintain momentum, surface blockers early, and adapt to emerging insights. The regular cadence of planning, reviews, and retrospectives helped align the team, improve transparency, and ensure that progress was both continuous and meaningful. It also facilitated clearer ownership of user stories and more realistic effort estimation, especially as our familiarity with the stack and project scope evolved.

We followed Scrum closely from the outset. Our product backlog was populated based on user stories tied to functional requirements, and we prioritized them using a mix of Bang For The Buck (BFTB) score and development dependencies. During sprint planning, we assessed the logical build order of features (e.g., implementing authentication before watchlist access) and aligned on sprint scope accordingly. We held weekly standups to track individual progress and unblock one another, and we concluded each sprint with a review and retrospective to evaluate outcomes and refine our team's workflow. This process helped us stay focused, adapt efficiently, and steadily build toward our Minimum Viable Product.

### 5.1 Product Backlog

| S.N<br>o. | User Story                                                                                                                                          | Priority | Estimated<br>Effort<br>(hrs) | Status    | Status Date | BFTB<br>Score |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------|-----------|-------------|---------------|
| 1         | As a new user, I want to create an account so I can save my preferences and receive tailored movie recommendations.                                 | High     | 10                           | Completed | 26/03/2025  | 1.00          |
| 2         | As a user, I want to log in securely so I can access my personalized content.                                                                       | High     | 8                            | Completed | 26/03/2025  | 1.25          |
| 3         | As a user, I want to log out of my account to protect my personal data and privacy.                                                                 | High     | 4                            | Completed | 09/04/2025  | 2.50          |
| 4         | As a user, I want to reset my password if I forget it, so I can regain access to my account.                                                        | High     | 6                            | Completed | 26/03/2025  | 1.67          |
| 5         | As a new user, I want to complete an onboarding questionnaire to input my genre preferences and interests, so I receive relevant movie suggestions. | High     | 10                           | Completed | 26/03/2025  | 1.00          |
| 6         | As a user, I want to view and update my movie preferences in the Profile section to receive more accurate recommendations.                          | Medium   | 8                            | Completed | 09/04/2025  | 0.75          |

|    |                                                                                                                                                          |        |   |           |            |      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|-----------|------------|------|
| 7  | As a user, I want to search for movies using keywords so I can quickly find titles I'm interested in.                                                    | Medium | 7 | Completed | 09/04/2025 | 0.86 |
| 8  | As a guest, I want to search and explore movies without signing up so I can get a feel of the app before creating an account.                            | Medium | 7 | Completed | 09/04/2025 | 0.86 |
| 9  | As a user, I want to see popular and trending movie suggestions so I can discover what others are watching.                                              | High   | 8 | Completed | 09/04/2025 | 1.25 |
| 10 | As a user, I want to save movies to a watchlist so I can revisit them easily later.                                                                      | Medium | 5 | Completed | 26/03/2025 | 1.20 |
| 11 | As a user, I want my login session to persist as I navigate through the app, so I don't need to re-authenticate on every page.                           | Medium | 9 | Completed | 09/04/2025 | 0.67 |
| 12 | As a user, I want to navigate smoothly across different views (Dashboard, Guest, Watchlist, Recents, etc.), so I can access various features of the app. | Medium | 6 | Completed | 23/04/2025 | 1.00 |
| 13 | As a user, I want to change my password securely from the Settings tab to maintain account security.                                                     | High   | 7 | Completed | 23/04/2025 | 1.43 |
| 14 | As a user, I want to limit the number of search results returned so the list is manageable and relevant.                                                 | Medium | 4 | Completed | 09/04/2025 | 1.50 |
| 15 | As a user, I want to filter movies by genre, release year, or rating so I can find movies that match specific criteria.                                  | Medium | 8 | Completed | 23/04/2025 | 0.75 |
| 16 | As a user, I want to remove movies from my watchlist once I've watched or lost interest in them.                                                         | Medium | 4 | Completed | 09/04/2025 | 1.50 |
| 17 | As a user, I want to view my recent activity so I can keep track of what I've searched or watched.                                                       | High   | 5 | Completed | 09/04/2025 | 2.00 |
| 18 | As a user, I want the filtering and sorting of movie lists (e.g., by rating or Title) to work properly, so I can explore content in meaningful ways.     | Medium | 6 | Completed | 09/04/2025 | 1.00 |
| 19 | As a user, I want to access a "Contact Us" page to provide feedback or request support when needed.                                                      | Medium | 4 | Completed | 23/04/2025 | 1.50 |
| 20 | As a user, I want to access the "About Us" page easily from the navigation menu, so I can find it when I need to know more about                         | Medium | 4 | Completed | 23/04/2025 | 1.50 |

|    |                                                                                                                                                                 |      |    |           |            |      |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-----------|------------|------|
|    | the app.                                                                                                                                                        |      |    |           |            |      |
| 21 | As a user, I want to click on a movie from the search or recommendation results to view its detailed information, so I can learn more about it before watching. | High | 10 | Completed | 23/04/2025 | 1.00 |
| 22 | As a user, I want to track my watchlist and see which streaming platforms host those movies.                                                                    | High | 5  | Completed | 09/04/2025 | 0.50 |
| 23 | As a user, I want my user details saved securely and correctly and be able to update them, so that I can enjoy the benefits of all features.                    | High | 8  | Completed | 23/04/2025 | 1.25 |

To help determine which user stories to prioritize, we first assigned the Business Value for each user story, following a structured approach based on the assigned priority levels in our backlog. Stories marked with High priority were given a Business Value of 10, Medium priority stories were assigned a value of 6, and Low priority stories received a value of 2. This quantification allowed us to objectively compare the importance of each feature. We then calculated the BFTB (Bang For The Buck) score by dividing the assigned business value by the estimated effort (in hours) required to complete each story. This ratio helped us identify which features provided the highest return for the development effort invested, enabling more informed decisions when prioritizing features during sprints.

## 5.2 Sprint 1 Details

When deciding which user stories to address first, we considered both logical dependencies and the calculated BFTB (Bang For The Buck) score. Stories that formed the foundation of the application — such as user authentication, onboarding, and preference saving — were prioritized early on because they enabled many other features to function correctly. Once these core components were in place, we used the BFTB scores to guide our development priorities, focusing on stories that offered the highest value relative to the effort required. This approach ensured that we built the application in a logical, scalable order while also maximizing impact within each sprint.

Below are the details of Sprint 1, as part of the demonstration of how we followed SCRUM closely throughout the development of the application.

### 5.2.1 Sprint 1 Planning

Sprint 1 Goal: Develop the foundational structure of the web app and implement core features to get basic functionality up and running.

### 5.2.2 Sprint 1 Product Backlog

| S.No. | User Story                                                                                                                                          | Priority | Estimate | Status      | Acceptance Criteria                                                                                             | Assigned To |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-------------|-----------------------------------------------------------------------------------------------------------------|-------------|
| 1     | As a new user, I want to create an account so I can save my preferences and receive tailored movie recommendations.                                 | High     | 10       | Completed   | User can create an account using a registration form with valid credentials, and data is stored in the backend. | Hong Wei    |
| 2     | As a user, I want to log in securely so I can access my personalized content.                                                                       | High     | 8        | Completed   | User can log in with correct credentials                                                                        | Hong Wei    |
| 3     | As a user, I want to log out of my account to protect my personal data and privacy.                                                                 | High     | 4        | In Progress | User can click a logout button and be securely logged out, returning to the homepage.                           | Hong Wei    |
| 4     | As a user, I want to reset my password if I forget it, so I can regain access to my account.                                                        | High     | 6        | Completed   | User can request a password reset via email and successfully update their password through a reset link.        | Hong Wei    |
| 5     | As a new user, I want to complete an onboarding questionnaire to input my genre preferences and interests, so I receive relevant movie suggestions. | High     | 10       | Completed   | The onboarding questionnaire has been designed and the component                                                | Bryan       |

|    |                                                                                                                               |        |   |             |                                                                                                                                      |           |
|----|-------------------------------------------------------------------------------------------------------------------------------|--------|---|-------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
|    |                                                                                                                               |        |   |             | is created, but it is not yet connected to backend logic or user flow.                                                               |           |
| 6  | As a user, I want to view and update my movie preferences in the Profile section to receive more accurate recommendations.    | Medium | 8 | In Progress | User can access a preferences page and update genres or interests, which are saved and used for recommendations.                     | Bryan     |
| 7  | As a user, I want to search for movies using keywords so I can quickly find titles I'm interested in.                         | High   | 7 | In Progress | User can input search keywords and receive a list of movie results from the API, though output may still be uncleaned or unfiltered. | Jeffrey   |
| 8  | As a guest, I want to search and explore movies without signing up so I can get a feel of the app before creating an account. | High   | 7 | In Progress | Guest users can view PopularCarousel and TopRated components on the guest page;                                                      | Han Lin   |
| 9  | As a user, I want to see popular and trending movie suggestions so I can discover what others are watching.                   | High   | 8 | In Progress | PopularCarousel and TopRated components display relevant movies on the dashboard using data from the API.                            | Han Lin   |
| 10 | As a user, I want to save movies to a watchlist so I can revisit them easily later.                                           | High   | 5 | Completed   | The watchlist UI is implemented and displays a                                                                                       | Hong Quan |

|  |  |  |  |  |                                                                   |  |
|--|--|--|--|--|-------------------------------------------------------------------|--|
|  |  |  |  |  | list of saved movies, with data linked to a dummy Firebase setup. |  |
|--|--|--|--|--|-------------------------------------------------------------------|--|

### 5.2.3 Sprint 1 Weekly Standup

|              |                            | 19th March                                                                                                                                                              | 26th March                                                                                                                                                                                           |
|--------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tai Hong Wei | What did you do last week  | I completed the Register and Login feature and implemented additional validation features to improve user input handling.                                               | I completed the forget password feature as well as created the logout component                                                                                                                      |
|              | What will you do this week | I will implement the Logout feature as well as the 'forget password' feature to allow users to change their password.                                                   | I will work on the user session to ensure that user's login state remains as he/she traverse through the app                                                                                         |
|              | Any challenges/obstacles   | It was my first time using Firebase Auth and I had quite a lot of obstacles coding out the functions and reading the documentation to understand the underlying concept | I have some problems trying to implement logout component to topbar, it doesn't appear and is hidden for some reason. There could be some misalignment between my component and the Topbar component |
| Bryan Ng     | What did you do last week  | I completed creating linking the user's preferences from the onboarding questions to firebase                                                                           | I added more questions needed to understand our users better                                                                                                                                         |
|              | What will you do this week | I will continue updating the user interface and add new questions to the onboarding page                                                                                | I will create a Profile page where users can change their movie preferences and a Settings tab for users to change their passwords                                                                   |
|              | Any challenges/obstacles   | I have some trouble with the designing of the web page                                                                                                                  | I had issues with the UI of the Questionnaire component as well as merging my code with others onto GitHub                                                                                           |

|                |                            |                                                                                                                                                           |                                                                                                                                                                        |
|----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jeffrey Peh    | What did you do last week  | I created the API function call for the search feature                                                                                                    | I linked the Topbar component to pass data using query parameters                                                                                                      |
|                | What will you do this week | I will work on linking the topbar to pass data using query parameters                                                                                     | I will work on the SearchResult component after users searches for a movie and design the UI                                                                           |
|                | Any challenges/obstacles   | I tried out the search API but search results gave a rather large JSON output back, I need to find a way to limit the results                             | I struggle with aligning the movie images using CSS as all of them comes in different sizes                                                                            |
| Li Han Lin     | What did you do last week  | I completed the component TasteReel which recommends movies based on users' saved preferences                                                             | I added two new components, PopularCarousel and TopRated which shows the users popular and top rated movies                                                            |
|                | What will you do this week | I will start on the PopularCarousel and TopRated components which are going to be part of the Guest and Dashboard page                                    | I will integrate the components to the Dashboard and Guest pages and work on their UI next                                                                             |
|                | Any challenges/obstacles   | I struggled with the styling of the components as the UI of the page doesn't seem to appear like how it is intended to be                                 | I can't seem to get the components' UI to work like how I want them to be, I had issues with CSS and swiper                                                            |
| Heng Hong Quan | What did you do last week  | I completed 2 components, Topbar and Sidebar which are the components that users will use to navigate throughout the app                                  | I completed the core functionalities of the Watchlist and StreamingDistribution component as well as the filter function but by using hardcoded movie data for testing |
|                | What will you do this week | I will work on the Watchlist and StreamingDistribution components as well as the filter function within Watchlist to allow users to filter through movies | Fine tune the UI for Watchlist page and create Recents page. Start to prepare interaction between Watchlist, Recents and Firebase                                      |

|  |                          |                                                                                                                      |                                                                                                                       |
|--|--------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|  | Any challenges/obstacles | I had problems with the CSS styling as it doesn't work well with other components and changed its size for no reason | I had issues with making the UI as what we designed it to be and also coding out the functions to link it to firebase |
|--|--------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|

## 5.2.4 Sprint 1 Review

|   |          |                                                                                                                                                                   |
|---|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Done     | User can successfully sign up with valid credentials                                                                                                              |
| 2 | Done     | User is able to log in using correct credentials and is redirected to their personalized dashboard.                                                               |
| 3 | Not Done | User is still unable to log out due to UI issues                                                                                                                  |
| 4 | Done     | User can request a password reset link and reset their password via email successfully.                                                                           |
| 5 | Done     | Questionnaire is up during onboarding and preferences are saved.                                                                                                  |
| 6 | Not Done | User can yet to modify their preferences after saving their preferences during onboarding.                                                                        |
| 7 | Not Done | User is able to input keywords and receive a list of relevant movie results; however, results are not yet filtered or cleaned for user-facing display             |
| 8 | Not Done | The UI for the Guest page, including components like PopularCarousel and TopRated, is implemented but not yet connected to the API or displaying real movie data. |
| 9 | Not Done | The PopularCarousel and TopRated components have been designed and added to the UI, but they are not yet fetching or displaying real movie data from the API.     |

|    |      |                                                                                                                  |
|----|------|------------------------------------------------------------------------------------------------------------------|
| 10 | Done | The watchlist UI is implemented and displays a list of saved movies, with data linked to a dummy Firebase setup. |
|----|------|------------------------------------------------------------------------------------------------------------------|

Feedback: While not all stories were completed, the sprint goal of setting up the web app skeleton and getting basic features running has been largely met, setting a strong foundation for Sprint 2

### 5.2.5 Sprint 1 Retrospective

| What went well                                                                                   | What didn't go well                                                                                                                                         | What have you learned                                                                                                     | What could be done better                                                   |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Completed key features like Register/Login, TasteReel, Watchlist, and Search functionalities.    | UI implementation didn't always match intended designs, causing repeated debugging and CSS inconsistencies between one another affected component rendering | Gained better understanding of Vuejs's component interactions as well as passing data between parent and child components | Establish shared design system to ensure consistent UI styling              |
| Helped each other debug issues with component alignment and styling.                             | Struggles with Git merging and pull requests resulting in major conflicts which slowed down integration                                                     | Improved knowledge of Firestore and Firebase Auth in Firebase                                                             | Do code reviews more frequently to catch misalignments early                |
| Arranged consistent weekly meetings at a certain time which improves efficiency and productivity | Firebase syncing sometimes caused delays due to inconsistent data structures                                                                                | Improved proficiency in using Git collaboratively (resolving merge conflicts, making pull requests etc)                   | Strengthen branching strategy and commit discipline to reduce merge issues. |

## 6 Tech Stack

To support a modern, responsive, and scalable movie recommendation web application, we adopted a carefully curated tech stack consisting of Vue.js, Firebase, and Git, among other tools and libraries. These technologies allowed us to build a feature-rich platform that supports real-time updates, seamless API integration, and a highly interactive user experience.

### 6.1 Frontend - Vue.js

Vue.js served as the core frontend framework due to its reactive, component-based architecture and developer-friendly syntax. It enabled us to build modular components such as `TopBar.vue`, `TasteReel.vue`, and `Watchlist.vue`, ensuring maintainability and reusability. Vue Router facilitated seamless navigation between views (e.g., Dashboard, Profile, Watchlist) without full page reloads, supporting a single-page application structure. Vue's reactivity system also allowed dynamic rendering based on user state and data updates, critical for real-time features like personalized recommendations.

### 6.2 Backend - Firebase

We used Firebase as a backend-as-a-service (BaaS) to handle authentication and database operations.

- Firebase Authentication was used to implement secure login, registration, logout, and password reset flows.
- Cloud Firestore (a NoSQL document database) was used to persist user preferences, watchlists, recents, and settings. Each user is stored as a document in the Users collection, with subcollections like Watchlist and Recents storing movie metadata.
- Real-time syncing ensured that any updates (like toggling a movie in the watchlist) were immediately reflected across the UI without manual refreshes.

### 6.3 External API – The Movie Database (TMDB) via Axios

To retrieve up-to-date movie information, we integrated with the TMDB API. All API interactions were handled using Axios, a promise-based HTTP client that simplified asynchronous requests and error handling. Axios was used to fetch movie metadata such as titles, genres, ratings, posters, and available streaming platforms. For personalized recommendations (e.g., in `TasteReel.vue`), we used user preferences to

construct API queries with specific genre and language filters, returning top-rated films sorted by TMDB's rating metrics.

## 6.4 Additional Libraries & Tools

- Swiper.js was used to implement the horizontal scrolling carousel for movie displays like the Popular Carousel. It offered mobile-friendly swipe gestures and built-in pagination.
- Vite was used as the build tool, offering fast development server startup and optimized bundling for production.
- SCSS was used for scoped component styling, allowing us to maintain a consistent dark theme with responsive design elements across devices.

## 7 Version Control & Collaboration – Git & GitHub

To facilitate collaborative development across a five-member team, we adopted Git as our version control system and GitHub as our central repository host.

We followed a feature branching strategy, where each developer worked on their assigned components in individual branches (e.g., `feature/profile`, `feature/guest_route`) and submitted pull requests (PRs) to the main branch. This allowed us to parallelize development while isolating features and minimizing merge conflicts.

Each PR underwent manual reviews before merging. Team members provided feedback, resolved merge issues, and discussed architectural decisions using GitHub's comment and issue tracking system. This transparent workflow improved code quality and enabled team-wide alignment on UI behavior, database schema, and component interactions.

During weekly sprint planning, tasks were distributed based on the product backlog and assigned to members using GitHub Issues and Projects. Standups and retrospectives allowed us to reflect on Git workflow bottlenecks, such as conflicting styles or inconsistent component naming, which we mitigated by standardizing file naming conventions and agreeing on a shared design system.

Throughout development, we adopted a disciplined Git workflow to ensure smooth collaboration across the team. Developers regularly committed their changes with clear, descriptive messages to make the version history easy to follow and debug. All work was carried out on individual feature branches, which helped isolate development efforts and reduce the likelihood of conflicts. Once features were ready, team members submitted pull requests to the main branch on GitHub, where code was reviewed before merging.

To prevent accidental overwrites or instability, we protected the main branch by requiring pull requests for all changes. Synchronization across local development environments was maintained through regular pulls from the main branch (`git pull origin main`), ensuring everyone was working with the most recent codebase. Although we faced some merge conflicts in earlier sprints, we became more proficient at resolving them through improved communication and structured commits. This Git-based workflow allowed us to develop in parallel, maintain traceability of changes, and integrate each team member's contributions into a cohesive application.

This structured Git workflow ensured our development remained stable, traceable, and collaborative, even as different features were built simultaneously.

## 8 Link to GitHub Repository

[https://github.com/BT3103AppDev1/I2-finaltermproject-2425s2I2\\_group\\_16](https://github.com/BT3103AppDev1/I2-finaltermproject-2425s2I2_group_16)

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